

AUCKLAND, New Zealand

WMO: 931100

Lat: 37.0089S

Lon: 174.8072E

Elev: 7

StdP: 101.24

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 4.1        | 5.3 | 1.9                           | 4.3 | 6.2  | 2.9 | 4.7 | 7.1  | 14.0                  | 12.3 | 12.5 | 12.7 | 1.6                   | 160  | 0.524 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 2             | 7.0                       | 25.8            | 20.1 | 24.8 | 19.6 | 23.9 | 19.0 | 21.5                | 23.9 | 20.7 | 23.1 | 20.1 | 22.5 | 5.7                     | 30   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.8                            | 15.5 | 22.7 | 20.0 | 14.7 | 22.0 | 19.1 | 13.9 | 21.5 | 62.5          | 23.6 | 59.8 | 23.2 | 57.5 | 22.7 |                   |

Extreme Annual Design Conditions

| Extreme Annual WS |      |                    | Extreme Annual Temperature |           |      |            | n-Year Return Period Values of Extreme Temperature |            |      |            |      |     |      |     |      |  |
|-------------------|------|--------------------|----------------------------|-----------|------|------------|----------------------------------------------------|------------|------|------------|------|-----|------|-----|------|--|
| Mean              |      | Standard Deviation |                            | n=5 years |      | n=10 years |                                                    | n=20 years |      | n=50 years |      |     |      |     |      |  |
| Min               | Max  | Min                | Max                        | Min       | Max  | Min        | Max                                                | Min        | Max  | Min        | Max  |     |      |     |      |  |
| (a)               | (b)  | (c)                |                            | (e)       | (f)  | (g)        | (h)                                                | (i)        | (j)  | (k)        | (l)  | (m) | (n)  | (o) | (p)  |  |
| 12.4              | 11.0 | 9.8                | DB                         | 2.0       | 27.7 | 0.6        | 1.1                                                | 1.6        | 28.5 | 1.2        | 29.1 | 0.8 | 29.7 | 0.4 | 30.5 |  |
|                   |      |                    | WB                         | 1.5       | 22.8 | 0.6        | 0.9                                                | 1.1        | 23.4 | 0.7        | 23.9 | 0.4 | 24.4 | 0.0 | 25.1 |  |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.6   | 19.9  | 20.4  | 19.0  | 16.7  | 14.3  | 12.0  | 11.2  | 11.7  | 13.1  | 14.4  | 16.0  | 18.4 |
|                                                                                  | DBStd       | 3.66   | 1.82  | 1.63  | 1.72  | 1.98  | 2.21  | 2.31  | 2.05  | 1.73  | 1.76  | 1.63  | 1.74  | 1.82 |
|                                                                                  | HDD10.0     | 28     | 0     | 0     | 0     | 0     | 1     | 8     | 12    | 6     | 1     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 1194   | 7     | 2     | 13    | 57    | 125   | 189   | 222   | 207   | 157   | 122   | 72    | 22   |
|                                                                                  | CDD10.0     | 2059   | 306   | 292   | 279   | 201   | 135   | 69    | 49    | 57    | 94    | 137   | 181   | 259  |
|                                                                                  | CDD18.3     | 182    | 54    | 61    | 34    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 3     | 23   |
|                                                                                  | CDH23.3     | 298    | 112   | 124   | 34    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 25   |
| CDH26.7                                                                          | 7           | 3      | 4     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |      |
| Wind                                                                             | WSAvg       | 4.7    | 4.8   | 4.4   | 4.3   | 4.0   | 4.2   | 4.2   | 4.3   | 4.5   | 5.0   | 5.7   | 5.5   | 5.2  |
| Precipitation                                                                    | PrecAvg     | 1140   | 72    | 76    | 81    | 94    | 102   | 121   | 125   | 118   | 99    | 81    | 82    | 89   |
|                                                                                  | PrecMax     | 1483   | 201   | 232   | 194   | 224   | 191   | 275   | 198   | 218   | 229   | 200   | 233   | 202  |
|                                                                                  | PrecMin     | 819    | 4     | 5     | 13    | 13    | 43    | 60    | 36    | 35    | 47    | 17    | 10    | 21   |
|                                                                                  | PrecStd     | 189    | 54    | 63    | 49    | 43    | 40    | 48    | 41    | 48    | 34    | 43    | 43    | 40   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.0  | 27.1  | 25.2  | 23.2  | 20.9  | 18.2  | 17.1  | 17.2  | 19.1  | 20.4  | 23.0  | 25.2 |
|                                                                                  |             | MCWB   | 20.4  | 21.2  | 19.3  | 19.0  | 17.9  | 16.3  | 14.8  | 14.2  | 15.2  | 16.0  | 18.3  | 19.5 |
|                                                                                  | 2%          | DB     | 25.8  | 25.9  | 24.2  | 22.0  | 19.8  | 17.1  | 16.1  | 16.2  | 18.0  | 19.2  | 21.3  | 24.0 |
|                                                                                  |             | MCWB   | 20.0  | 20.3  | 18.9  | 18.3  | 17.1  | 15.2  | 13.9  | 13.5  | 14.8  | 15.3  | 17.1  | 18.9 |
|                                                                                  | 5%          | DB     | 24.8  | 24.9  | 23.2  | 21.1  | 18.8  | 16.2  | 15.2  | 15.5  | 17.1  | 18.2  | 20.2  | 22.9 |
|                                                                                  |             | MCWB   | 19.5  | 19.8  | 18.4  | 17.7  | 16.5  | 14.4  | 13.1  | 13.0  | 14.2  | 14.9  | 16.3  | 18.3 |
|                                                                                  | 10%         | DB     | 23.7  | 23.9  | 22.2  | 20.1  | 17.9  | 15.8  | 14.6  | 14.9  | 16.1  | 17.3  | 19.2  | 21.9 |
|                                                                                  |             | MCWB   | 18.9  | 19.2  | 18.1  | 17.1  | 15.8  | 14.0  | 12.6  | 12.5  | 13.7  | 14.6  | 15.7  | 17.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 22.3  | 22.7  | 21.3  | 20.7  | 18.7  | 17.2  | 15.6  | 15.4  | 16.5  | 17.1  | 19.1  | 21.1 |
|                                                                                  |             | MCDB   | 24.8  | 25.0  | 23.0  | 22.0  | 19.9  | 18.1  | 16.6  | 16.5  | 18.1  | 19.0  | 21.5  | 23.4 |
|                                                                                  | 2%          | WB     | 21.2  | 21.8  | 20.4  | 19.5  | 17.7  | 15.7  | 14.5  | 14.3  | 15.5  | 16.3  | 18.1  | 20.1 |
|                                                                                  |             | MCDB   | 24.0  | 24.1  | 22.5  | 21.0  | 19.0  | 16.7  | 15.7  | 15.7  | 17.1  | 18.2  | 20.3  | 22.2 |
|                                                                                  | 5%          | WB     | 20.5  | 21.0  | 19.6  | 18.6  | 16.9  | 15.0  | 13.6  | 13.6  | 14.9  | 15.6  | 17.3  | 19.4 |
|                                                                                  |             | MCDB   | 23.1  | 23.4  | 22.0  | 20.3  | 18.3  | 16.1  | 15.0  | 15.1  | 16.5  | 17.5  | 19.4  | 21.7 |
|                                                                                  | 10%         | WB     | 19.6  | 20.3  | 18.8  | 17.7  | 16.0  | 14.1  | 12.9  | 12.9  | 14.1  | 15.1  | 16.5  | 18.6 |
|                                                                                  |             | MCDB   | 22.3  | 22.8  | 21.5  | 19.6  | 17.6  | 15.4  | 14.3  | 14.5  | 15.9  | 17.1  | 18.8  | 21.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.0   | 7.0   | 7.1   | 6.8   | 6.3   | 6.1   | 6.2   | 6.2   | 6.2   | 6.0   | 6.3   | 6.3  |
|                                                                                  |             | MCDBR  | 8.2   | 8.0   | 7.8   | 7.2   | 6.5   | 5.8   | 6.2   | 6.6   | 6.7   | 6.9   | 7.4   | 7.7  |
|                                                                                  |             | MCWBR  | 4.0   | 3.9   | 3.9   | 4.2   | 4.4   | 4.5   | 4.6   | 4.6   | 4.5   | 4.5   | 4.5   | 4.1  |
|                                                                                  | 5% WB       | MCDBR  | 6.7   | 6.5   | 6.5   | 5.7   | 5.4   | 5.3   | 5.5   | 5.8   | 5.9   | 6.1   | 6.5   | 6.2  |
|                                                                                  |             | MCWBR  | 4.0   | 3.8   | 3.9   | 4.0   | 4.1   | 4.6   | 4.5   | 4.4   | 4.3   | 4.5   | 4.6   | 4.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.351  | 0.351 | 0.329 | 0.318 | 0.312 | 0.308 | 0.303 | 0.313 | 0.334 | 0.342 | 0.338 | 0.349 |      |
|                                                                                  | taud        | 2.546  | 2.568 | 2.633 | 2.653 | 2.639 | 2.629 | 2.640 | 2.606 | 2.524 | 2.507 | 2.544 | 2.555 |      |
|                                                                                  | Ebn at Noon | 983    | 960   | 944   | 896   | 843   | 816   | 844   | 888   | 922   | 958   | 990   | 989   |      |
|                                                                                  | Edh at Noon | 108    | 101   | 88    | 76    | 67    | 63    | 66    | 78    | 96    | 107   | 108   | 108   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 6.70   | 5.80  | 4.85  | 3.44  | 2.38  | 1.91  | 2.13  | 2.93  | 3.99  | 5.15  | 6.08  | 6.36  |      |
|                                                                                  | RadStd      | 0.34   | 0.42  | 0.27  | 0.21  | 0.15  | 0.14  | 0.14  | 0.17  | 0.23  | 0.32  | 0.33  | 0.49  |      |

Historical Trends

|      | DBAvg                  | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (2 neighbors) | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -77     | +99     | N/A     |

Nomenclature: See separate page